From Semantic Seed to Object Mask: Controlled Selection of Diffusion Attention for Training-Free Segmentation

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Abstract

Text-to-image diffusion models contain semantic localization in cross-attention and spatial grouping in self-attention. Turning these signals into masks raises a practical question: which observations and refinements help beyond a strong training-free reference, and which merely fit evaluation images? We study class-word segmentation on 490 VOC-Sim images with five class-stratified folds and transfer each frozen choice to 707 COCO-Cap images. A controlled progression retains cross-attention at 8, 16 and 64 pixels, propagates its seed through one 64-pixel self-attention module, applies two observation-wise passes, and refines the mask with a fixed DenseCRF. It reaches 0.813 out-of-fold mean IoU on VOC-Sim and 0.637 on COCO-Cap. Most of 45 registered comparisons fail: selective graph operators do not help, and observation-reliability operators that help one-pass propagation substitute for the second pass and are absorbed by DenseCRF. Selecting on the refined mask instead finds one surviving change, SA calls 10–20, reaching 0.816 out-of-fold and 0.644 on COCO-Cap. We analyze why a second pass completes objects while later arrivals are increasingly false positives. A separate exploratory attention-weighted seed reaches 0.823 in a forced out-of-fold comparison and 0.662 on COCO-Cap; we distinguish this from the registered selection path. Published OVAM and other baselines are reported under their original evaluation conditions. The results show that extraction, propagation and refinement must be selected jointly, with a clear boundary between validated gains and promising follow-up observations.

1. Introduction

Diffusion models generate an image while repeatedly connecting text tokens to image positions and image positions to one another. Cross-attention (CA) offers a semantic seed for a queried class; self-attention (SA) can carry that evidence across the object's extent. DAAM aggregates CA for attribution [10], while OVAM extends attribution to words absent from the generation prompt and studies SA fusion and CRF refinement [6]. This makes generated-image segmentation possible without fitting a segmentation network.

The difficulty is selecting *which* attention to use. Stable Diffusion 1.5 exposes 31 denoising calls, four spatial scales, sixteen CA and sixteen SA modules, and eight heads per module, and the attention graph can be propagated, filtered or re-weighted in many ways. A combination tuned on the images used to choose it may look better than it is, and a restriction that helps alone may stop helping once another component changes. We therefore treat the design as a sequence of questions, each answered on five VOC-Sim folds and only then transferred, frozen, to COCO-Cap.

The sequence has three roles (Fig. 1). CA must find a semantic core of the queried class; SA must grow that core into the object without leaking into the background; and an image-aware refinement must sharpen the boundary. The answers led to a simple training-free pipeline: average CA over the non-32-pixel modules, propagate the seed through the last 64-pixel SA module twice within each call–head observation, and apply OVAM's fixed DenseCRF. It scores 0.813 out-of-fold (OOF) mean IoU on VOC-Sim and 0.637 on COCO-Cap.

The route there is part of the result (Fig. 3). Of 45 registered comparisons, one per experiment family and stage, most did not survive the fold rule, and several plausible graph operators were harmful. A family of observation-reliability operators did improve one-pass propagation, by about one point, yet none improved on a second pass and none survived DenseCRF: they were substitutes for object completion, not additions to it. This pattern changed how we selected. Later studies chose settings *after* refinement, which uncovered the one change that survives the full pipeline: restricting SA to calls 10–20, selected in four of five folds (0.816 OOF, 0.644 COCO-Cap). Later CA-weighting and multi-scale studies raise COCO-Cap transfer but are not selected on VOC-Sim; we report them as measured extensions with explicit selection status.

The paper contributes a traceable selection protocol for

CA, SA and post-processing; a staged analysis of semantic seeding, object completion and leakage, with the negative results that shaped it; and a published-baseline comparison that separates regenerated-image results from the original paper's results.

2. Related work

Attention-based segmentation. DAAM aggregates diffusion CA to attribute image regions to prompt words [10]. OVAM introduces a separate attribution prompt, allowing a queried word that need not occur in the generation prompt, and evaluates attention-based pseudo-masks with SA and DenseCRF post-processing [6]. Our target remains the queried class word throughout. We study which available attention observations to aggregate and how to propagate the resulting seed, using the same two generated benchmark families.

Spatial propagation and refinement. SA defines an affinity between latent image positions. Repeated application of a row-stochastic affinity spreads a CA seed and can fill missing object pixels, but can also cross an object boundary. We therefore evaluate pass count, SA scale and temporal window together with the final mask. DenseCRF uses image appearance to refine boundaries [3]; we retain the fixed OVAM parameterization. Recent training-free diffusion segmentation work also studies attention structure and geometry [7, 9]. NERVE combines semantic-output head weighting, local affinity, and a stopped random walk in a CLIP–diffusion setting [5]; our separate SG29 follow-up tests adaptations to a single class-word seed and the original SD softmax SA. SPARK sharpens diffusion-derived affinity through expansion, inflation, pruning, and normalization for unsupervised segmentation [4]; our SG30 follow-up isolates these graph operations while retaining a class-word CA seed. SeeDiff iteratively extracts new seeds between SA scales and refines foreground with propagated background evidence [8]; our SG31 follow-up tests these mechanisms using the frozen class-word CA source. These follow-ups are not substituted for the confirmed pipeline.

Hierarchical residuals and Markov reachability. FA-Seg combines refinement from several SA resolutions using fixed weights [1]. M2N2 views SA as a Markov transition, uses first-arrival time as a spatial signal, and tests temperature sharpening and iterative proportional fitting [2]. Our SG33 follow-up retains the sequential coarse-to-fine hierarchy: earlier states enter only as residual support, and class-seeded Markov reachability replaces only its coarse initialization. Neither branch reproduces the cited method; both remain experiments until the frozen VOC selection and COCO transfer are complete.

Optimized tokens. OVAM also learns attribution tokens from annotated support images [6]. A separate experiment in this repository reaches higher scores with those released tokens, but that uses mask supervision. This paper's main method and comparison use ordinary class words; optimized-token results appear only as a distinct analysis, never as training-free evidence.

3. Method

Semantic seed. For an image generated by Stable Diffusion 1.5 and target class word w, we query the text encoder with a separate attribution prompt, including every tokenizer piece of w. At each denoising call, CA module and head, the conditional image-query/text-key softmax yields a spatial token map. We average the word-piece maps and selected observations after bilinear resizing to 64×64. The selected CA set uses all 31 recorded calls, all eight heads and eleven modules at resolutions 8, 16 and 64; it excludes the five 32-pixel modules. The aggregate is normalized per image to $[0, 1]$ to form c. This class-word seed requires no learned token.

Observation-wise propagation. For SA observation $i = (t, h)$ in the last 64-pixel module, let $A_i \in \mathbb{R}^{4096\times4096}$ be its row-stochastic query–key affinity. One and two passes are

$$p_1 = \frac{1}{|\mathcal{I}|}\sum_{i\in\mathcal{I}} A_i c, \qquad p_2 = \frac{1}{|\mathcal{I}|}\sum_{i\in\mathcal{I}} A_i^2 c. \tag{1}$$

The square is taken within each observation before averaging; $\left(\frac{1}{|\mathcal{I}|}\sum_i A_i\right)^2 c$ is a different operator. All SA heads are retained. The confirmed system uses all 31 calls; a later joint study selects calls 10–20, inclusive, with the rest unchanged.

Mask and boundary refinement. We min–max normalize the propagated map, upsample it bilinearly to annotation resolution, and threshold it with a threshold selected on VOC training folds. The fixed OVAM DenseCRF implementation then takes this binary prior and the generated RGB image; we do not tune its parameters on COCO-Cap. We call this refinement R2. No ground-truth mask enters map extraction, propagation or inference.

Exploratory CA weighting. A later study weights CA modules using a comparison between within-module query/key affinities and a paired SA affinity. Its A3 variant uses a transposed resized SA affinity in the implemented similarity score. We report exactly this implemented operator and its measured output; it is not an exact implementation of the published GoCA score [7]. Its selection

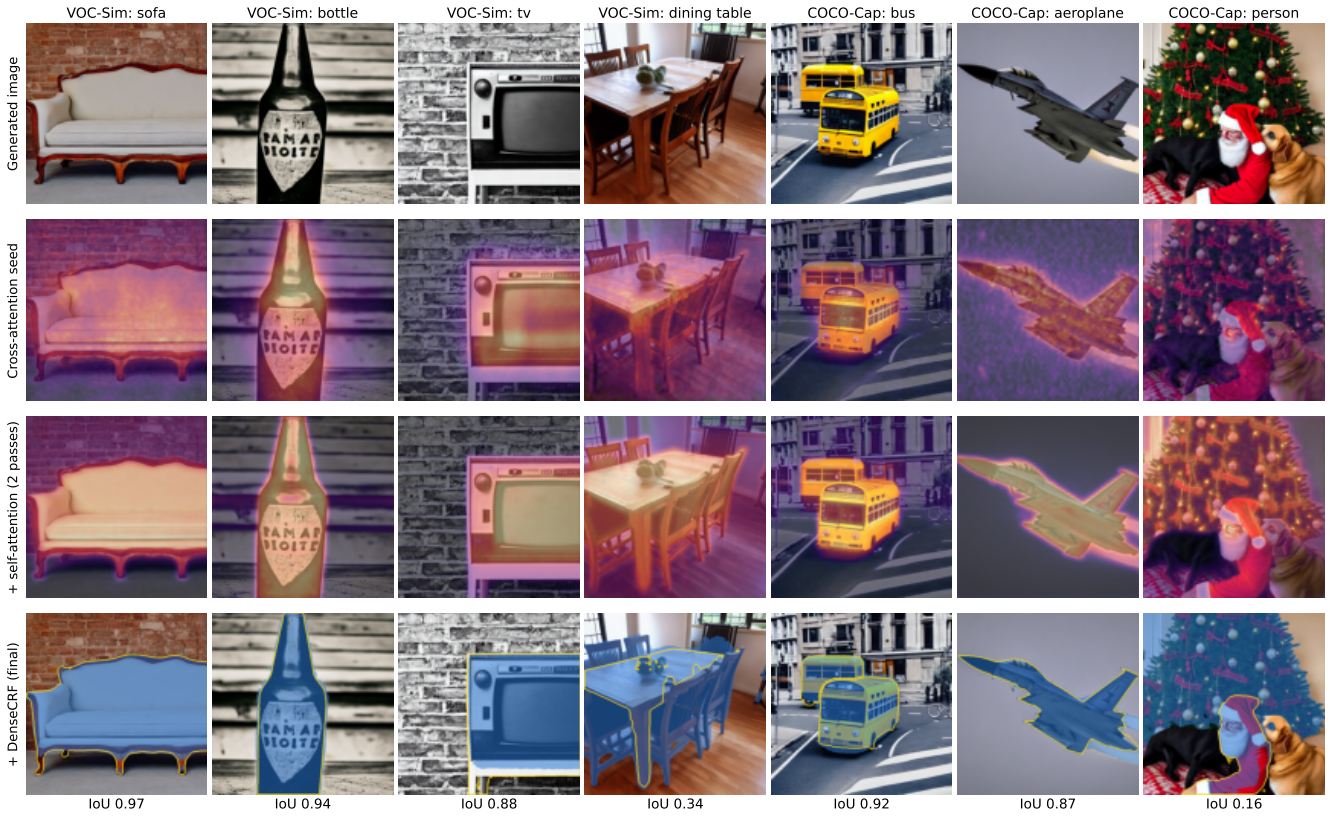


Figure 1. **What each stage of the training-free pipeline contributes.** Rows: the generated image; the cross-attention seed c of the class word; the map after two self-attention passes, $\mathrm{mean}_i(A_i^2 c)$; the final mask after DenseCRF at the frozen threshold 0.45 (blue), with the ground-truth outline (yellow) and the image IoU. Columns are chosen by a fixed rule, without inspecting any image: the classes at the best, upper-third, lower-third and worst class IoU on VOC-Sim, and at the best, median and worst on COCO-Cap, each shown by the image whose IoU is the median of its class. The last column of each dataset is therefore a failure case: the dining-table mask spreads from the tabletop onto the chairs, and the weak person seed is propagated into the Christmas tree behind the figure.

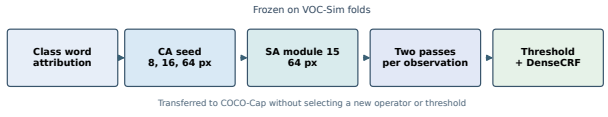


Figure 2. **From semantic evidence to an object mask.** A class-word attribution prompt yields a CA seed; the selected 64-pixel SA module propagates that seed twice within each call–head observation; the observations are averaged and the fixed DenseCRF refines the thresholded map using RGB boundaries. We select settings on VOC-Sim folds and only then score COCO-Cap. The middle-call variant changes only the SA observation set.

and software-audit status are given separately in Sec. 8 and Sec. 9.

4. Experimental protocol

Data and metric. We evaluate binary masks for a queried VOC class on the 490 annotated VOC-Sim images and 707 annotated COCO-Cap images released with OVAM [6]. We regenerate the images from released prompts and seeds to record the attention needed by our method; generation details can produce visual differences from the released renders. VOC-Sim is the selection set. COCO-Cap is evaluated only after the operator and threshold are frozen. We report class IoU from pooled TP, FP and FN within each class and the unweighted mean of 20 class IoUs. This is different from averaging per-image IoUs.

Fold selection. The 490 VOC-Sim images are assigned once to five class-stratified folds of 98 images. For each held-out fold, a candidate is chosen on the other four folds and scored on the held-out fold. The resulting OOF score estimates the selection procedure. We also identify *forced OOF* scores for a named candidate when it is evaluated on all held-out folds even when the registered selection rule would not choose it. We never conflate forced OOF with a promoted method.

Early one-axis screens use a fixed 0.45 mask threshold,

with 0.35 and 0.55 as calibration checks. A candidate must improve the training-fold mIoU by at least 0.010, or by at least 0.005 with 12 of 20 classes improving and no class losing more than 0.05. Differences within 0.002 favor the more inclusive setting. Promotion requires at least four of five fold votes and a positive OOF change. Promoted settings are tested jointly before a stage is frozen. Later factorial studies select their mask threshold from a registered 17-point VOC grid within each training fold; full-VOC threshold selection is used only to freeze COCO transfer. This distinction matters when comparing scores from different study phases.

Comparisons and uncertainty. We report each stage against its same-run reference. Paired image-IoU intervals use 20,000 bootstrap resamples; these intervals complement, rather than replace, the class-macro mIoU and fold decision. Published OVAM benchmark numbers are shown separately from our regenerated-image scores. The released baseline masks in our repository are also scored separately, because some methods used different generated images or modified evaluation code.

How the results are organized. Each of Secs. 5–8 asks one question about the pipeline, states what was tried, gives the fold evidence and the decision, and says which question the decision raised next. Figure 3 summarizes the whole search before the details: one point per registered experiment family, against the reference of its own run. Panel (a) shows operators on the propagated map and panel (b) complete pipelines with DenseCRF. The horizontal position is an in-sample full-VOC change, used only to place the point; whether a family was promoted is the out-of-fold decision of its registered rule.

5. Semantic localization: selecting the CA seed

Question: does CA give the queried class a reliable semantic core? A CA map should identify that core before any SA expansion. We first hold propagation aside and screen denoising calls, resolutions, modules, heads and attribution conventions. This separates a weak seed from an overly aggressive graph operation.

Spatial scale matters more than a preferred head. Removing the five 32×32 CA modules raises OOF mIoU from 0.665 to 0.691 in the combined CA confirmation; four folds choose it. The paired mean-image IoU change is $+0.022$ (95% bootstrap interval $[+0.014, +0.029]$). The effect persists at 0.35, 0.45 and 0.55, although average precision falls from 0.922 to 0.913: the chosen seed is more selective at the operating threshold, not a universal improvement in pixel ranking. An example in the supplementary material shows why scale is a substantive choice: the example's 32-pixel map highlights only a narrow part of the airplane, while the retained scales offer complementary extent and detail. This is an illustration, not evidence that every 32-pixel layer is poor.

On separate screens, removing CA head 1 and some module subsets appeared useful. Once crossed with the resolution choice, the extra head restriction changes mIoU by only $+0.0004$ and lies inside the 0.002 tie band. We therefore keep all eight heads. The conditional-query attribution convention, with the softmax over 77 text positions and the mean of all pieces of the class word, also survives; the tested OVAM-style unconditional/first-piece alternative is 0.016 lower in its own screen. Denoising-call restriction is not retained in this initial CA stage.

Decision and next question. We freeze the seed: eleven CA modules at 8, 16 and 64 pixels, all calls and heads, conditional queries and the mean over word pieces. A later test of contrastive seeds (SG5), which compare the class-word attention with the start-of-text, non-target or padding positions, use OVAM's two-key normalization, or average fixed synonyms, was retained by the reference in all five folds, both for the raw seed and inside the final pipeline; its best variant lost 2.7 COCO-Cap points (Fig. 3a). The frozen seed is selective but incomplete (Fig. 1, second row): it marks the most class-specific parts of the object and leaves the rest weak. That raises the next question, how to complete the object from this core.

6. Object completion: selecting SA propagation

Question: how should SA grow the core into the object? A semantic core can be accurate yet incomplete. We therefore treat SA as a propagation operator, not as another class detector. One pass through all sixteen SA modules improves the frozen seed in the SA module screen, but the last 64-pixel module (`up_blocks.3.attentions.2`, index 15) is better: OOF mIoU rises from 0.756 to 0.789 in the combined SA confirmation, and all five folds select this module. Its paired mean-image IoU gain is $+0.042$ with interval $[+0.037, +0.047]$. Eighteen classes improve; person loses 0.055, an early warning that diffusion can also erase or misassign object extent.

Separate screens favored call 30, resolutions 8+64 and heads $\{1, 4, 6\}$, but none survives the SA combination. With module 15 fixed, the head subset lowers mIoU by 0.002 and call 30 lowers it further. A Cartesian audit of six CA seeds and twelve SA settings plus no-SA controls retains the sequential CA/SA pair in all five folds. Module 15 is the best SA partner for every audited CA seed. This rules out the simple explanation that the sequential choice

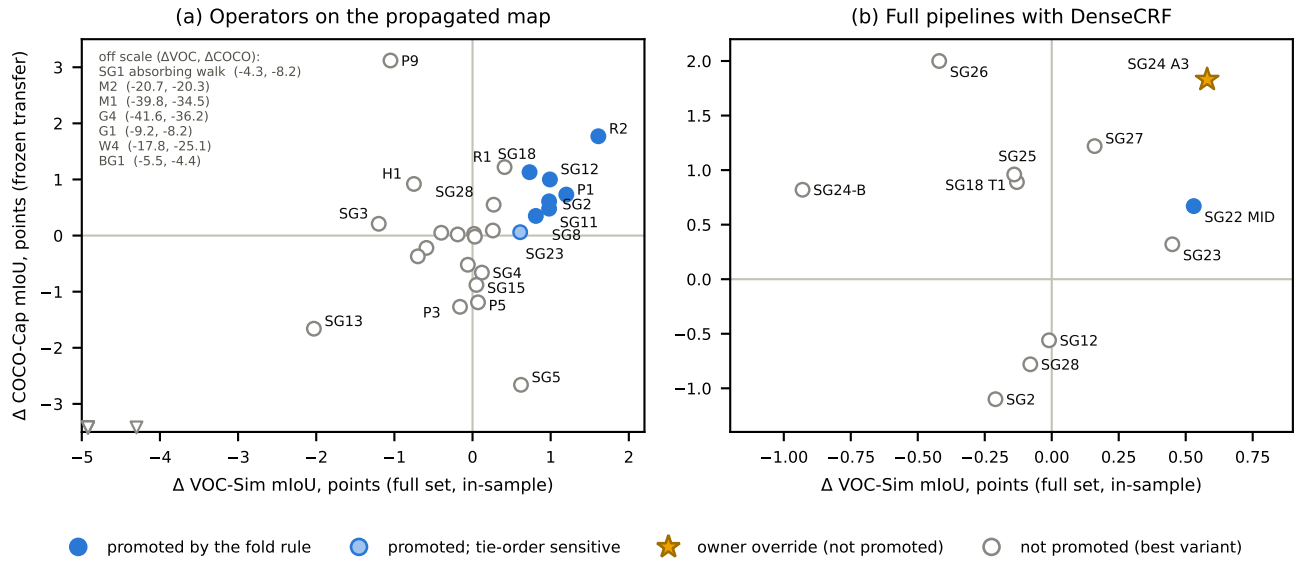


Figure 3. **The search at a glance.** Each point is one registered experiment family, compared with the reference cell of the same run (same generated images). x: change in VOC-Sim mIoU at each cell's full-VOC threshold (in-sample, descriptive); y: change in frozen COCO-Cap transfer. The plotted cell is the one the fold rule promoted, or else the family's best variant. (a) Operators on the propagated map, mostly against one SA pass. The promoted cluster (blue) gains about one point on both datasets; apart from the second pass and DenseCRF, the promoted operators that were carried to the next stage (SG2, SG12, SG18) do not improve on two passes plus DenseCRF (panel b). (b) Complete pipelines with DenseCRF, against the final pipeline of the same run. Only the middle-call window (SG22) is promoted; the A3 CA weighting (star) was retained by owner decision after the registered selector kept its reference. Several studies (SG25–SG27) raise COCO-Cap transfer but are not selected on VOC-Sim. Per-family values and protocol sections are listed in the supplementary material.

worked only because the CA seed happened to be frozen first.

How far should evidence propagate? With that same module, adding a second observation-wise pass raises OOF mIoU from 0.788 to 0.800 (+0.012); three passes are not promoted. The second pass particularly helps chair (+0.078) and person (+0.030). With the fixed DenseCRF added, the combined confirmation selects two passes plus R2 in all five folds and reaches 0.813 OOF, compared with 0.788 for one unrefined pass. Table 1 lists these same-study stages, without implying that the independent CA and SA screens are additive.

The mechanism is a completion–leakage tradeoff (Fig. 4). In the diagnostic decomposition, pass two raises VOC recall from 0.912 to 0.924 at nearly unchanged precision; its largest mean image-IoU gain is in the lowest third of seed coverage (+0.029). A third pass adds pixels that are increasingly often wrong. A first-arrival diagnostic, recorded later with the SG25 package (A3 seed, calls 10–20, threshold 0.425), finds that pixels first accepted at depth one are 86.5% true positive on VOC and 70.8% on COCO, while pixels first accepted at depth three are only 39.3% and 14.0%. Further propagation therefore has diminishing reliability, especially under COCO clutter. These are diagnostic pixel statistics, not an alternative segmentation score.

Can a better graph replace plain propagation? If later arrivals cause the leakage, an operator that propagates more selectively should help. We registered and ran twenty-four families of alternatives against the one-pass reference (Fig. 3a). Graph-restricting operators did not survive. An absorbing walk between confident foreground and background (SG1), confidence-locked propagation (SG3), reciprocal links (SG9), a seed-connected bottleneck (SG10), SA region labelling (SG4, SG13) and fourteen backlog operators, from DiffSeg-style merging to CA-weighted prototypes and seeded region growing, and none was promoted; several lost tens of points. The bottleneck score, for example, did not separate added true from false pixels (AUC 0.35–0.56).

One family did help: operators that weight or filter the 248 call–head observations by their agreement. Edge consensus (SG2), sign consensus (SG8), robust trimmed aggregation (SG11) and keeping the 50% most consistent observations (SG12) each raised one-pass OOF mIoU from 0.787 to 0.797–0.798 and were promoted. This is, however, the level that a second pass already reaches (0.800). Applied on top of two passes, SG2 was retained by the plain two-pass reference in all five folds. Their per-class gains also

resemble those of the second pass (class correlation 0.88 for SG2 and 0.82 for SG12; supplementary material). Observation reliability and a second pass are thus substitutes, two routes to object completion, not complementary gains. A separate temporal screen (SG18) promoted a middle-call window, calls 10–20, in all five folds.

Decision and next question. We freeze SA module 15, all calls and heads, and two observation-wise passes. The promoted alternatives were then carried to the full pipeline, which raised the next question: does a gain on the propagated map survive refinement?

7. Boundary refinement and joint temporal selection

Question: what does refinement add, and does a gain on the propagated map survive it? Propagation changes object extent; the final mask still needs a threshold and image-aware boundaries. We compared seed mixing, multiplicative gating, several normalizations and per-image Otsu calibration against the frozen propagation pair. Some calibration variants raise overall mIoU but violate the registered class-loss guard. The fixed DenseCRF is chosen in all five folds, improves all twenty classes in its screen, and raises OOF mIoU from 0.788 to 0.805 in that experiment. When two-pass propagation and DenseCRF are confirmed together, the combined system reaches 0.813 OOF and is selected in all five folds. The difference between the single-axis and combined gains is why we confirm interactions rather than adding isolated deltas.

Refinement absorbs propagation gains. We then carried the promoted propagation operators into the refined pipeline, each against two passes plus DenseCRF in the same run. None survived (Fig. 3b). SG2 plus DenseCRF scores 0.811 forced OOF against 0.813 and wins no fold; SG12 plus DenseCRF scores 0.812 and is retained by the reference in all five folds. The SG18 middle-call window, which had won all five raw folds, also loses after DenseCRF with one pass (0.811 versus 0.814, no fold), although it gains 0.9 COCO-Cap points. Gains measured on the propagated map are therefore not additive with refinement. The decision changed our procedure: from here on, every candidate is selected on the *refined* mask. It also left one cell untested, the middle-call window *with* two passes and DenseCRF.

Time and refinement interact. The factorial study SG22 fills that cell: it crosses SA calls 0–30 versus 10–20 with one, two and three propagation passes, before and after R2. On raw maps, the registered selector keeps all calls. After R2, it chooses calls 10–20 with two passes in four of five folds, and the selection path scores 0.8158 OOF against 0.8132 for the same-generation ALL-P2+R2 reference. Evaluated in every held-out fold, MID-P2+R2 scores 0.8168 forced OOF, and 0.6438 versus 0.6372 on frozen COCO-Cap. The result supports a specific sequence—select temporal evidence for the *refined* mask—rather than a general assertion that calls 21–30 are poor. In a separate diagnostic, late calls both recover some true pixels and add distant false positives; their utility varies by class and scene.

Setting	P2	R2	Mid	Forced OOF	Δ
One pass	–	–	–	.7873	–
Two passes	✓	–	–	.7997	+.0124
+ refinement	✓	✓	–	.8132	+.0259
+ calls 10–20	✓	✓	✓	.8168	+.0295

Table 1. **The nested system on VOC-Sim.** All four SG22 rows use the same frozen CA seed and SA module 15. P2 means two observation-wise passes; R2 is fixed DenseCRF; Mid keeps calls 10–20 rather than 0–30. OOF is the forced OOF mIoU of each cell (threshold chosen on the training folds); Δ is relative to the one-pass row. The registered selection path, which chooses MID-P2+R2 in four of five folds, scores 0.8158. The preceding CA and SA selection steps are reported separately in Sec. 5 and Sec. 6.

8. Transfer and published comparison

Question: does the VOC-selected system transfer, and does anything beyond it? We now freeze every choice on VOC-Sim and score COCO-Cap. The confirmed all-call training-free pipeline gives 0.637 mIoU; the later joint temporal selection gives 0.644. COCO-Cap includes captions in which the queried class name is absent, so separate class-word attribution is important. Neither the temporal window nor the threshold is chosen on COCO-Cap.

Re-scored released masks of the published methods, with their method-specific conditions, are in the artifact registry; they do not replace the published numbers.

Class-level transfer is weak. Aggregate gains transfer more reliably than their class pattern. Across fourteen within-run comparisons, the correlation between a change's per-class effect on VOC-Sim and on COCO-Cap ranges from -0.01 to $+0.46$ (supplementary material). The middle-call window, for instance, improves 16 VOC and 17 COCO classes, but VOC person loses (supplementary material). A class that a change helps on one benchmark is therefore weak evidence that it helps on the other.

Beyond the validated system. After SG22 the open question moved back to the seed and to SA scale; each study keeps the validated SA and DenseCRF. A CA module weighting (SG24-A) scores the agreement of each CA module's query–key affinity with a paired SA affinity. Its reg-

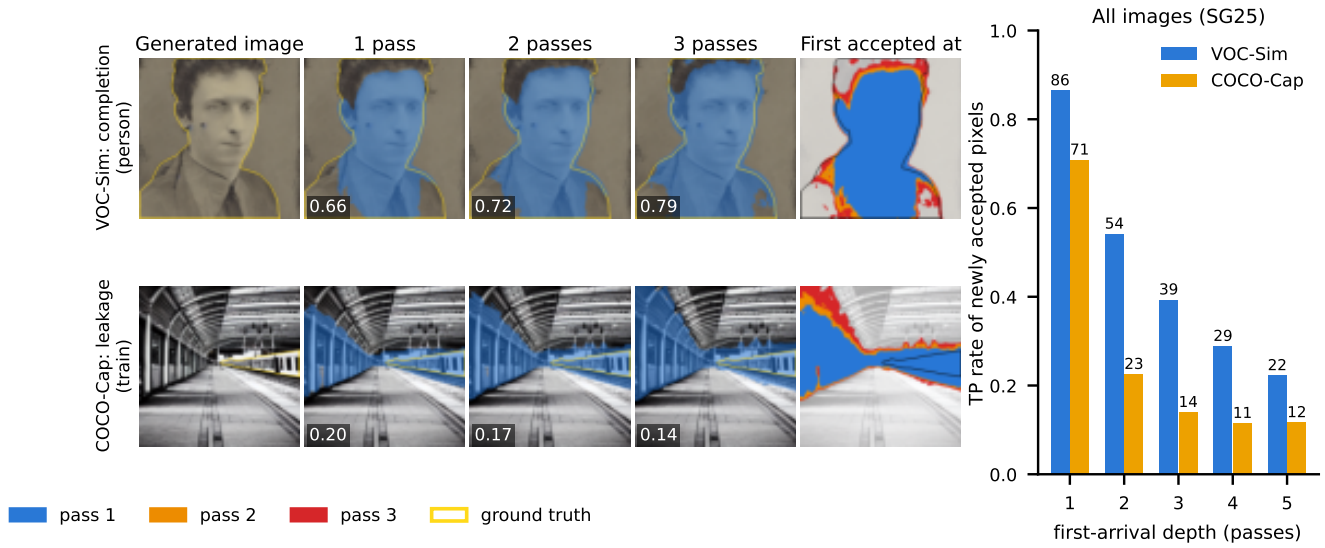


Figure 4. **How far should evidence propagate?** Left: two images chosen by a fixed rule from the all-call raw maps of the SG22 run, without inspecting any image: on VOC-Sim, the image at the 90th percentile of the two-pass gain (completion); on COCO-Cap, the image at the 10th percentile of the three-pass gain (leakage). The supplementary material shows both rules on both datasets. Columns: the generated image with the ground-truth outline; the raw mask after one, two and three observation-wise passes at threshold 0.45, with the image IoU; and the pass at which each accepted pixel was first accepted. Right: the true-positive rate of newly accepted pixels at each depth, pooled over all 490 VOC-Sim and 707 COCO-Cap images, from the SG25 diagnostic (A3 seed, calls 10–20, threshold 0.425; depths 1–5). The third pass still helps the completion example, but pooled over images most pixels it first accepts are false positives on both datasets.

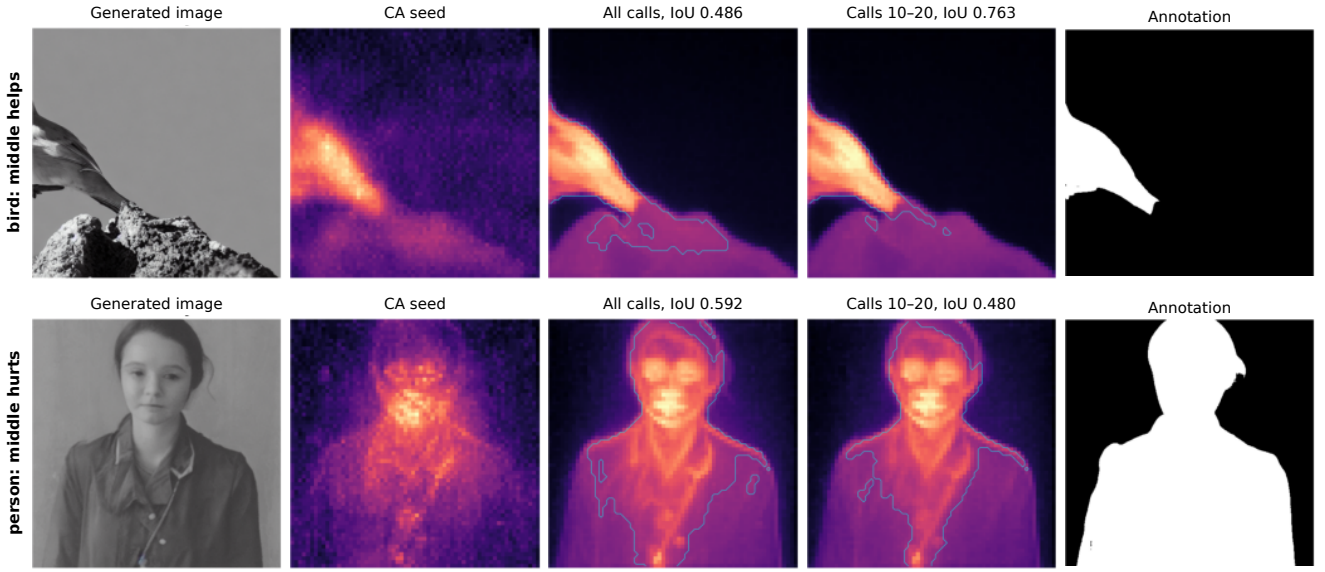


Figure 5. **A temporal window can remove leakage or remove object pixels.** Two VOC-Sim examples from the earlier SG18 one-pass, raw-map diagnostic: the same CA seed is propagated with all SA calls (0–30) or only calls 10–20. The blue contour is the predicted mask boundary; the original annotation is at right. Bird gains IoU when a false region is suppressed; person loses IoU when useful extent is suppressed. These are deliberately opposing examples of the mechanism, *not* images from the final two-pass-plus-R2 evaluation.

istered multi-candidate selector kept the unweighted reference; the A3 variant nonetheless scores 0.823 forced OOF versus 0.817 for the same-run reference, and 0.662 versus 0.644 on COCO-Cap, and was retained as the working default by owner decision. Its full-VOC threshold gives 0.824, which is in-sample and is not used as the main VOC es-

Method	VOC-Sim	COCO-Cap
Published in OVAM, Table 1		
DAAM	.662	.484
Attn2Mask	.687	.550
OVAM, class word	.704	.582
Published methods with extra training		
Grounded Diffusion	.621	.502
DatasetDM	.803	.593
Ours, regenerated images / released annotations		
All-call P2+R2	.813†	.637*
Middle-call P2+R2	.816†	.644*
A3 middle-call P2+R2‡	.823	.662*

Table 2. **Class-word results against the numbers reported by OVAM** [6]; optimized-token rows are deliberately excluded. OVAM Table 1 evaluates each baseline on its authors' renders and annotations, whereas our attention requires regeneration from prompts and seeds and uses the released annotations. Thus cross-block differences are indicative, not a paired head-to-head estimate. †VOC OOF of the registered selection path; *COCO with VOC-frozen threshold and method. ‡Exploratory forced-OOF A3 result; it was not the registered multi-candidate winner and uses the implemented transposed-affinity score.

timate. Restoring the five 32-pixel CA modules under A3 (SG24-B) loses 0.8 forced-OOF VOC points and wins no fold, while gaining 0.8 on COCO-Cap.

The studies that follow, all on the A3 seed, share one pattern: coarser SA helps transfer and hurts selection. Propagating through a 16-pixel SA module before the 64-pixel one (SG25) raises frozen COCO-Cap from 0.662 to 0.698 and improves 19 of 20 COCO classes, yet scores 0.818 VOC OOF against 0.823 and improves only 9 VOC classes. Coarse-guided SA (SG26) repeats this: after DenseCRF, each new cell loses 0.4–2.5 forced-OOF VOC points and gains 1.3–2.0 COCO-Cap points. A 46-formula late-fusion screen (SG27) retains the fine-scale reference in all five folds; its best COCO cell, 0.705, loses VOC and sits at the edge of the threshold grid. A PEARL-inspired study (SG28, [9]) found one boundary-aware Laplacian that gains 0.4 raw VOC points, but only $+0.0001$ after DenseCRF, so again the gain does not survive refinement. Later studies (SG29–SG33; supplementary Sec. G) also keep the reference after DenseCRF. These results suggest that COCO-Cap's cluttered scenes benefit from coarser context that the simpler VOC-Sim scenes do not reward. Under the registered rule, they do not change the selected system.

9. Discussion and limitations

Why the search took this route. Each decision opened the next question. A selective seed (Sec. 5) needed completion; completion by more propagation leaked, which motivated selective graph operators (Sec. 6); the operators that helped were substitutes for the second pass and were absorbed by DenseCRF, which moved selection to the refined mask (Sec. 7); and selection on the refined mask found the middle-call window. Throughout, components that pass an independent screen can vanish in combination: head and call subsets disappear once crossed with the module choice, and the middle-call gain appears only after DenseCRF. Most of the record in Fig. 3 is negative, and the negatives kept the final pipeline simple.

Where the pipeline fails. The fixed-rule columns of Fig. 1 show the two characteristic failures. When the queried object shares SA affinity with co-occurring objects, propagation fills them too: the dining-table mask covers the chairs. When the seed is weak, propagation amplifies whatever it does mark: the COCO person seed spreads into the Christmas tree behind the figure. The class pattern across fourteen within-run comparisons agrees. On VOC-Sim, person has the largest mean loss and is the most volatile class (seven gains, seven losses); potted plant and bicycle, with thin and porous structures, lose far more often than they gain (8 and 6 losses against 1 and 3 gains). On COCO-Cap, dining table loses in nine of fourteen.

What the figures do and do not show. All qualitative examples are chosen by a stated rule before inspection. The deliberately opposing examples in Fig. 5 show why a temporal restriction needs a cohort-level test. The first-arrival rates in Fig. 4 explain why deeper diffusion eventually leaks, but they come from a later package and do not prove a universal per-pixel confidence rule.

Selection and implementation boundaries. The main all-call pipeline and the later middle-call refinement were selected by their registered VOC rules. A3 received a favorable forced pairwise comparison after the multi-candidate selector kept A0; consequently it is exploratory, its full-VOC score is in-sample, and its gain holds only in the final pipeline (its CA-only mIoU falls). An independent code audit found that its resized SA affinity was transposed relative to the GoCA-inspired equation. On four images the correction changes per-call module weights by 0.9–2.0% (relative), so a large change is unlikely, but until a corrected replay is evaluated the A3 number describes only the implemented operator. SG25 and SG27 further reuse that seed and inherit this limitation.

The benchmarks contain images generated from prompts and seeds, not natural-image segmentation benchmarks. Our regenerated images may differ from the renders annotated by OVAM, and some published baselines used their own generation settings. The literature table is therefore contextual, not a paired comparison. All choices are made on VOC-Sim and one diffusion backbone; the 20-class macro mean also hides class and image-level variation. A

separate optimized-token track uses annotations and is not part of the training-free claim.

10. Conclusion

The most reliable training-free recipe in this study is a selective CA seed, observation-wise propagation through one fine SA module, two passes for object completion, and fixed image-aware refinement, with SA restricted to the middle calls once selection is made on the refined mask. Its failure mode is the tradeoff that shaped it: more propagation or coarser scale completes objects but also leaks.

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